7th International Conference on
Electronics and Sustainable Communication Systems
ICESC 2026
26-28, August 2026
https://icescs.com/2026/
icescsconf@gmail.com

# *Acceptance Letter*

**Paper ID:** ICESC_325

**Title:** Decision Tree Based Machine Learning Framework for Early Risk Prediction in Soldier Health Monitoring Using IoT and LoRa Communication

**Author(s):** M A Seshasaai, Sai Janavi P H, K Bala Manikanta, Kamaleshwaran M, Lakshana R, S Vijayalakshmi

---

*Decision: Accepted with Major Revision*

Herewith, the conference committee of the International Conference on Electronics and Sustainable Communication Systems [ICESC-2026] is pleased to inform you that the peer reviewed research paper entitled ***"Decision Tree Based Machine Learning Framework for Early Risk Prediction in Soldier Health Monitoring Using IoT and LoRa Communication"*** has been accepted for oral presentation as well as it will be recommended in ICESC Conference Proceedings. ICESC will be held on 26-28, August 2026, in Hindusthan Institute of Technology, Coimbatore, India. ICESC encourages only the active participation of highly qualified delegates to bring you various innovative research ideas.

*We congratulate you on being successfully selected for the presentation of your research work in our esteemed conference.*

Regards,

**Conference Chair,**
**ICESC**

Proceedings by

Technical Co-Sponsor

# *Review Comments*

## Decision: Acceptance with Major Revision

***Title –*** *Decision Tree Based Machine Learning Framework for Early Risk Prediction in Soldier Health Monitoring Using IoT and LoRa Communication*

---

### Review Comments 1:

1.The reported 99.56% accuracy may be misleading because real sensor readings are combined with logically and mathematically generated samples in the training dataset; the number of real and synthetic samples should be clearly reported by the authors, and the final model should be validated on an independent real-world dataset only.
2.A major inconsistency in model performance is present because only 85.09% mean accuracy is given by five-fold cross-validation with a very high 15.49% standard deviation, while 99.56% is reported as the final test accuracy; possible data leakage, overfitting, or improper data splitting should be investigated by the authors, and this large difference should be explained.
3.The claim of "early risk prediction" is made by the paper, but the current health condition is only classified by the Decision Tree using present sensor readings and a future health event is not actually predicted; a prediction horizon and time-based experiments should be included, or the claim should be changed to real-time health-risk classification.
4.The eight health classes such as Fatigue, Panic_Risk, Injury_Risk, Heat_Stroke, and Severe_Cold_Risk are not supported by clearly defined medical diagnostic criteria; exact labeling rules, clinical references, and expert validation for each health condition should be provided by the authors.
5.The dataset is highly imbalanced, with 2,461 Normal samples but only 231 Heat_Stroke and 250 Severe_Cold_Risk samples, yet class-wise performance is not properly discussed; precision, recall, F1-score, sensitivity, specificity, and confusion results for every class should be reported and analyzed separately.

### Review Comments 2:

1. Battery life analysis should be included.
2. Soldier comfort considerations need discussion.
3. Why choose the Decision Tree algorithm?
4. Practical deployment challenges need explanation.
5. LoRa packet loss needs evaluation.
6. Table 1 is not cited in the manuscript,Kindly add the corresponding in-text citation. Some minor English language editing is required, especially for spelling mistakes.